# <u>**Climate Policy and Economic Outcomes**</u>

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## <u>**Introduction:**</u>

Cities are major hubs for economic activity and energy use, and therefore play an important role in how we should address localized climate policy. In recent years, many U.S. cities have adopted climate policies with goals to reduce carbon emissions and promote clean energy usage. However, there is still debate about how these policies affect local economies. Some people argue that strong climate policies may slow economic growth or increase costs, while others believe they can actually support job creation and long-term city and economic development. For the undergraduate celebration of research, I examined the economic effects of the Paris Agreement (Global climate policy) at the national level across three countries. This report builds on that research by focusing specifically on U.S. cities, providing a more localized perspective on how climate policy is related to economic performance.

This report examines how climate policy commitments are related to economic outcomes in U.S. cities. My research question is: Do U.S. cities with stronger environmental policy commitments have different economic outcomes compared to cities with weaker commitments? To answer this question, I will focus on the following three indicators of economic well-being:

median household income, unemployment rate, and poverty rate. These variables provide a clear way to compare economic conditions across cities.

Major cities such as New York City and San Francisco have strong climate policies, while other cities have weaker policies. This report uses data from The American Council for an Energy-Efficient Economy (ACEEE) of 75 U.S. cities with different levels of climate policy strength. These different levels of strengths in climate policy allow for a fair analysis of differences in economic performance across the 75 cities.

However, it should be noted that cities with stronger climate policies tend to be larger and wealthier, which may influence their economic conditions and their ability to adopt these climate policies. Therefore, the results of this study should be interpreted as showing an association rather than a direct causal relationship.

Overall, the goal of this project is to better understand how climate policy is connected to economic conditions in urban areas. The results can be used to help inform policy makers and guide discussions about how cities can reach their environmental goals while also supporting and increasing local economic growth.

## Literature Review

Research on the economic outcomes of climate policy has shown some mixed results. Some studies show that climate policies can reduce emissions without really causing major harm to economic performance. The Organisation for Economic Co-operation and Development (OCED) finds that “environmental policies have little aggregate effect on economic outcomes ” and firms are still able to achieve their environmental goals (OECD, 2021). This suggests that having climate policies in place does not necessarily reduce employment, wages, or overall economic activity.

Other research has found economic benefits from climate policy. A report from the World *Unlocking the Inclusive Growth Story of the 21st Century: Accelerating Climate Action in Urgent Times* argues that climate action can support economic growth and job creation. For example, the report states that “Low-carbon growth could deliver economic benefits of US$26 trillion to 2030—and this is a conservative estimate… Taking ambitious climate action could generate over 65 million new low-carbon jobs in 2030” (page 12). This shows that investments in clean energy and sustainable infrastructure can create new economic opportunities, especially in sectors such as renewable energy and urban development (Mountford et al., 2018). This suggests that having better climate policies would increase long-term economic growth instead of stunting it.

At the same time, there is also evidence that economic conditions can influence which cities adopt stronger climate policies and which have weaker ones. Research on Climate Policy from the London School of Economics finds that changes in climate policy leads to shifts in economic structure, such as “In the short term, jobs will shift from high-carbon activities to low-carbon activities” and long-term effects “associated with innovation, job creation and growth” (Fankhauser et al., 2008). In addition, The 2024 City Clean Energy Scorecard ranks San Francisco, Seattle, Denver, Los Angeles, Oakland, Minneapolis, New York, Portland, San José, and Washington, DC as the top ten cities with the strongest clean energy policies. These cities are also known to be wealthy and have strong local economies, which may make it easier for them to adopt and support more ambitious climate policies.

Overall, research shows that the relationship between climate policy and economic outcomes is not always clear. Climate policies can create both costs and benefits, and their effects may vary across cities and over time (OECD, 2021; Fankhauser et al., 2008). This project builds on the research by focusing more specifically on U.S. cities and examining how differences in climate policy strength relate to differences in economic performance such as median income, unemployment, and poverty.

## Data

I use two different datasets to perform this analysis on the relationship between climate policy and economic performance in U.S. cities. The first one is from *The 2024 City Clean Energy Scorecard*, published by the American Council for an Energy-Efficient Economy (ACEEE). This dataset includes 75 U.S. cities and assigns each city a policy score based on the strength of its clean energy and climate policies. The scores are based on factors such as building efficiency, transportation policy, energy usage, and local government actions (Samarripas et. al, 2024). I went ahead and put the 75 cities along with their scores in a csv file in Excel, which is ready to be used in R for analysis. The policy scores are the main independent variable.

The second dataset I use is from the American Community Survey (ACS), collected by the U.S. Census Bureau. The ACS provides data based on 5 year estimates on economic and demographic conditions in the United States. I use the 2018-2022 estimates and focus on median household income (B19013), total population below the poverty line (B17001), and labor force and unemployment counts (B23025), which are used to calculate poverty and unemployment rates. Unemployment and poverty rates are calculated using ACS population counts for unemployed people, total labor force, and people below the poverty line. These are the dependent variables for this project and are used to measure economic performance across cities. The data is from the tidycensus package in R.

The two datasets were merged based on city and state names to create a combined dataset for this analysis. Since the ACS data is reported using geographic labels that do not exactly match city names in the ACEEE dataset, some data cleaning was required to ensure proper matching. In addition, not all cities matched perfectly. The ACEEE dataset includes 75 cities. After merging with ACS data, Boise, ID and Indianapolis, IN did not match successfully and were excluded because of missing ACS values. So the final regression uses 73 cities instead of 75.

There are some limitations to this data and the analysis. First, the ACEEE policy score measures the strength of climate policies, but not how effectively those policies are implemented. Second, ACS data is estimated based on survey data, so it may not be exact. It is also often reported at larger geographic levels, so it may not perfectly represent actual city-level economic conditions. Finally, differences in city factors such as size and population can influence policy strength and economic outcomes, which could affect the results of this analysis. The GitHub repository with all the R code used for this analysis is available at:
https://github.com/chriskor372/plan372-climate-policy-econ-project.git

## Methods

I used regression analysis to examine the relationship between climate policy strength and economic outcomes across U.S. cities. The independent variable is the ACEEE climate policy score for each city. The dependent variables are median household income, unemployment rate, and poverty rate, which are used to measure economic performance.

All analysis was done in R. After merging the ACEEE and ACS datasets, I created new variables to measure unemployment rate and poverty rate using ACS population counts (B01003). The final dataset includes 73 cities with complete data for all variables. I created three scatterplots to visualize the relationship between climate policy scores and each economic variable. A line of best fit was added to each plot to help show overall trends.

I then made three separate linear regression models using the climate policy score as the independent variable and one of the three economic outcomes as the dependent variable. These models estimate the average relationship between climate policy strength and economic outcomes across U.S. cities.

The results of this analysis should be interpreted as associations rather than causal effects because it compares different cities at one point in time rather than tracking changes over time. Other factors and differences such as city size and existing economic conditions can influence environmental policy strength and economic outcomes.

## **Results**

The analysis shows different relationships between climate policy scores and the three economic variables. The strongest relationship is the one between climate policy score and median household income (Figure 1). The regression results show that for each one-point increase in ACEEE policy score, median household income increases by about $351.05, and this relationship has a p-value below 0.001. The R-squared value is about 0.43, meaning that climate policy score explains about 43% of the variation in median household income across the cities included in this analysis (Figure 4). This shows that cities with stronger climate policies also tend to have higher median household incomes.

The relationship between climate policy score and unemployment rate is much weaker (Figure 2). The regression coefficient is negative, meaning that higher policy scores are associated with slightly lower unemployment rates. However, this result is not statistically significant because the p-value is about 0.4006 (Figure 5). This means there is not enough evidence here to prove that climate policy strength is directly related to unemployment rates.

The poverty rate model shows a clearer relationship (Figure 3). The regression results show that for each one-point increase in ACEEE policy score, poverty rate decreases by about 0.05215 percentage points. The p-value is 0.001467 so the result is statistically significant. However, the R-squared value is only about 0.1215 (Figure 6), meaning that policy score explains only about 12% of the variation in poverty rates. This shows that cities with stronger climate policies tend to have lower poverty rates, but it should be noted that many other factors also affect poverty.

In conclusion, the results suggest that stronger climate policy scores are associated with higher median household income and lower poverty rates, but do not show a clear pattern for unemployment rate. These findings do not necessarily prove that climate policies cause better economic outcomes, but rather show an association. One possible explanation is that wealthier cities have more resources to adopt stronger climate policies in the first place. Therefore, the results should be interpreted carefully.

## Policy Implications and Recommendations

The results of this analysis have important implications for urban policy. One implication is that cities should not avoid strong environmental policies out of concern that they will harm economic performance. Similar concerns have influenced climate policy decisions at the national level, such as the U.S. withdrawal from the Paris Agreement twice under both Trump administrations (“What Were the Main Reasons Cited by Trump for Leaving the Paris Agreement?”, 2025). However, this analysis suggests that these concerns may be overstated, which may also apply to cities. The results suggest that cities with stronger climate policies are not experiencing worse economic outcomes and may sometimes even have better outcomes.

The results should be interpreted with caution. Because this analysis shows associations rather than causal effects (as previously stated), we cannot definitively conclude that climate policies directly improve economic outcomes. It is likely that wealthier cities are better able to adopt strong climate policies due to their better financial resources and administrative capacity. Therefore, policymakers should consider existing economic conditions when making and implementing climate policies.

Based on the findings, one recommendation is for cities to invest in areas such as clean energy and public transportation. Cities such as New York City have implemented building efficiency policies that require large buildings to reduce energy use over time, which can lower energy costs while reducing emissions (“Local Law 97”, 2025). In addition, Seattle has invested in public transportation and clean mobility options. The Seattle Transportation Plan makes travel more reliable, sustainable, and affordable by investing in public transit and mobility infrastructure (Valencia & Lewis, 2024). These policies should be paired with efforts to support economic equity, especially in cities with fewer resources. Investing in job training and affordable energy programs can help distribute the benefits of climate policy more evenly.

Another recommendation is for future research to take a closer look at how climate policies actually work in cities and who they impact the most, especially across different communities and those of lower income. This could help identify which policies best support environmental and economic goals while ensuring that benefits are shared more evenly.

In conclusion, this analysis suggests that climate policy can play an important role in building more sustainable and equitable cities. While the results do not prove causation, they show that climate policies can exist alongside strong local economies and should continue to be considered by city governments as they may even help support long-term economic stability and quality of life for residents.

## **Appendix**

Figure 1: Climate Policy and Median Household Income Scatterplot

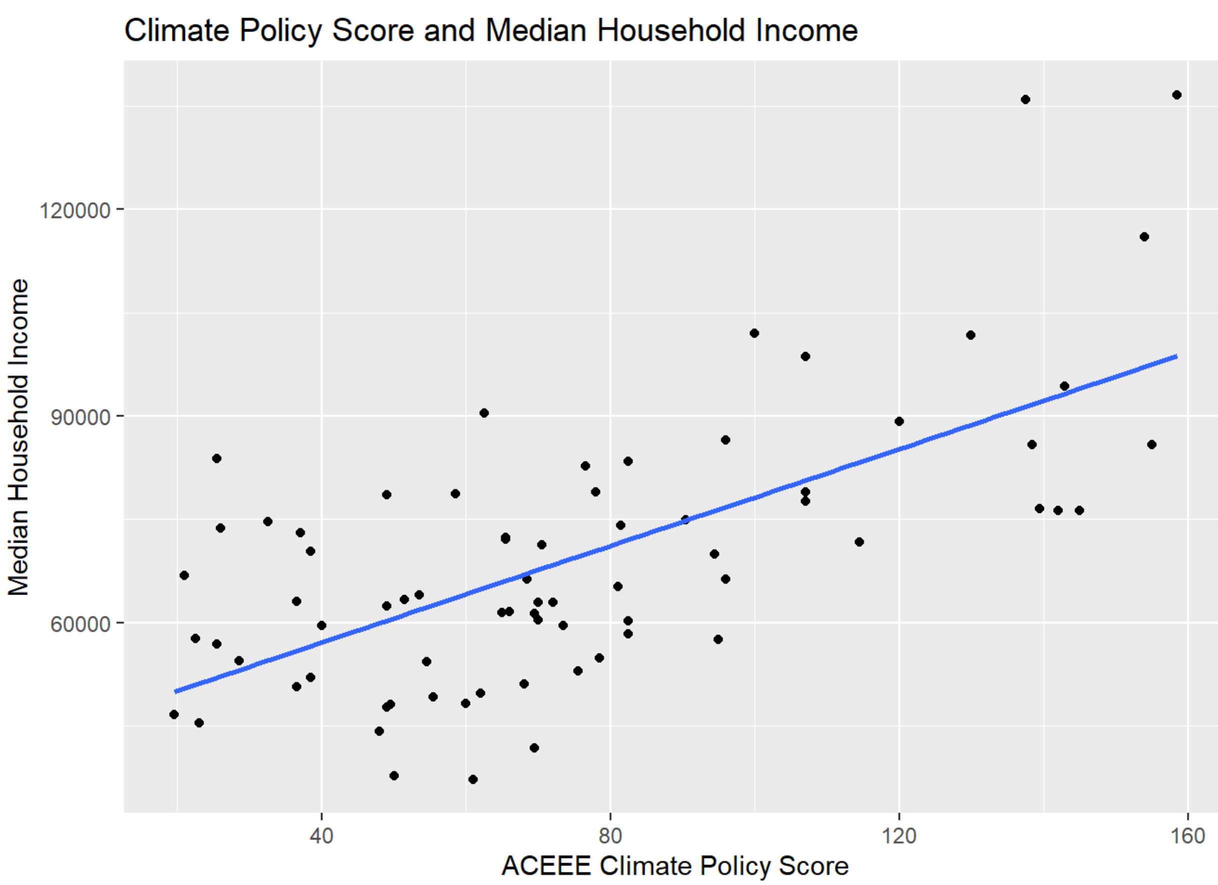

Figure 2: Climate Policy Score and Unemployment Rate Scatterplot

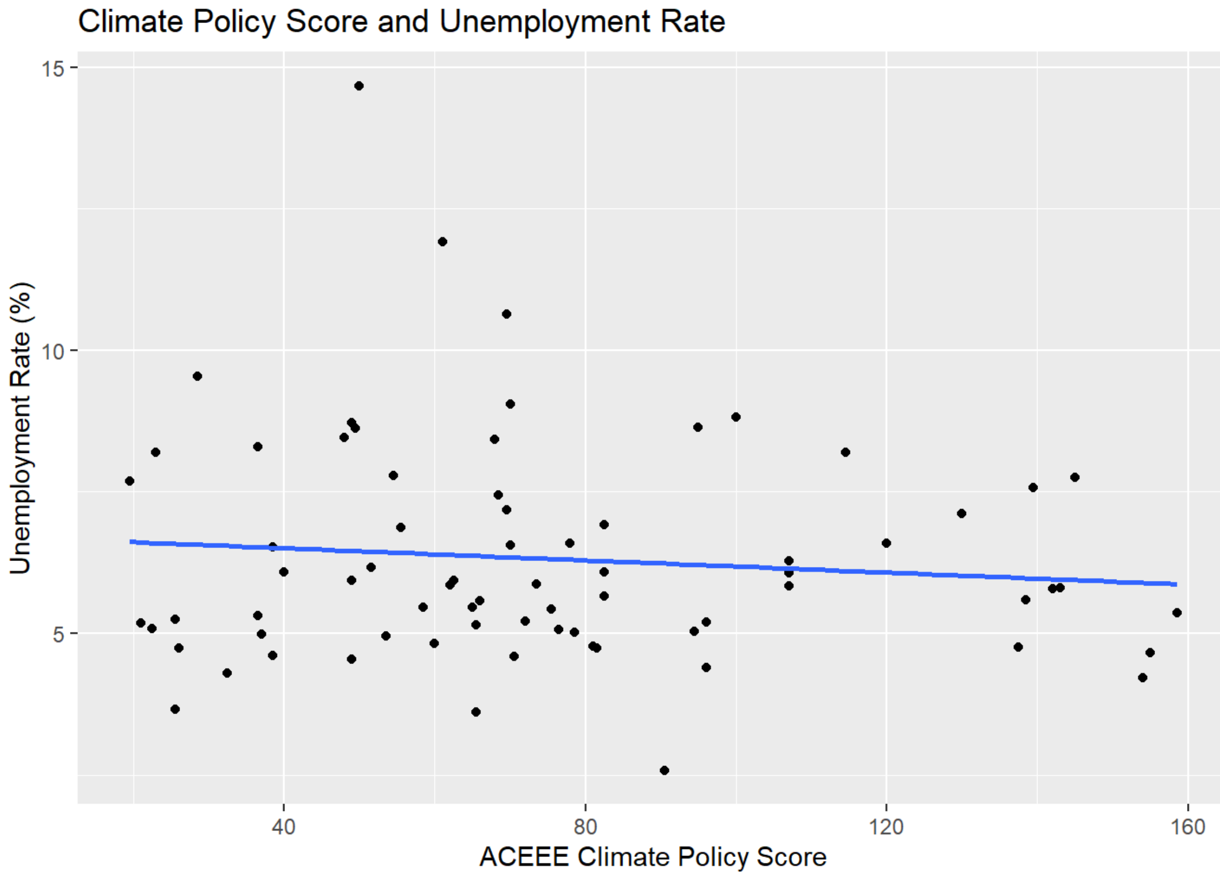

Figure 3: Climate Policy and Poverty Rate Scatterplot

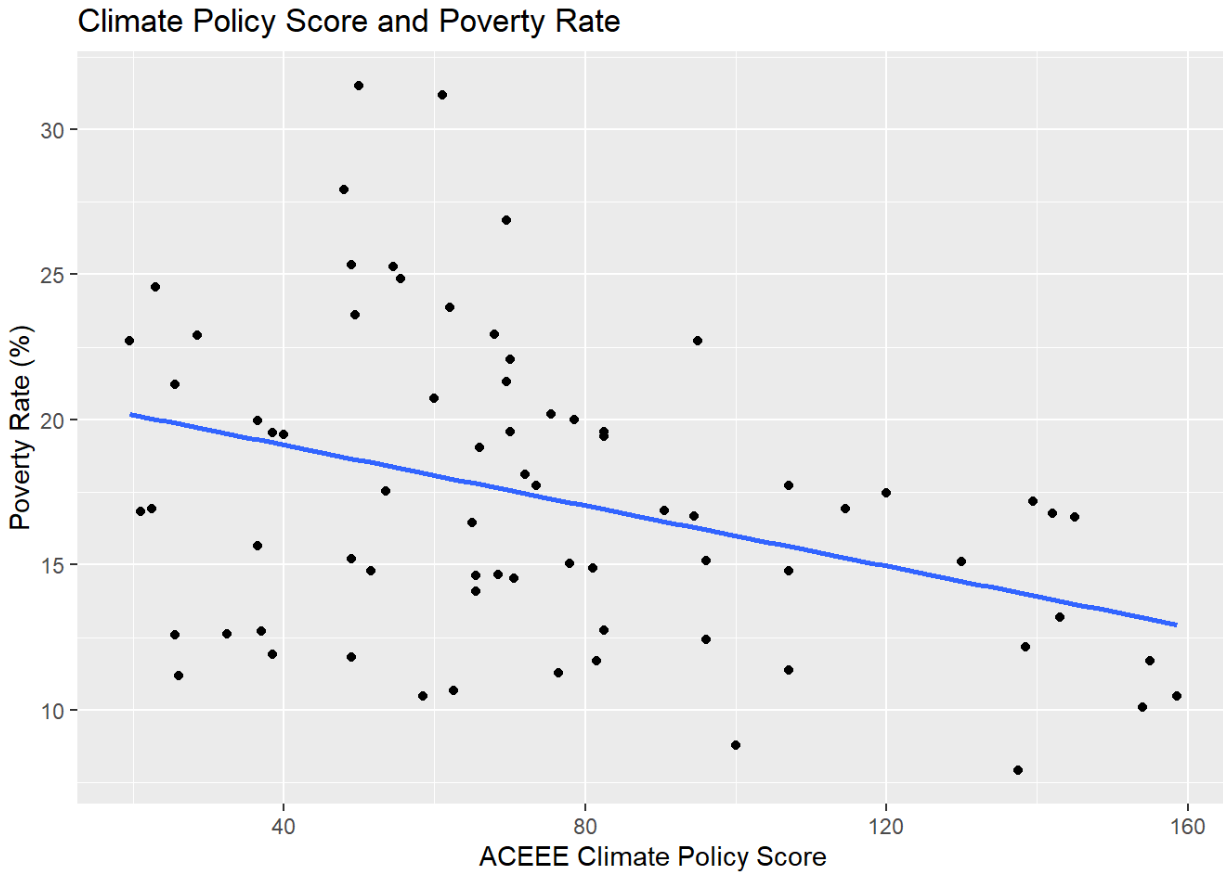


Figure 4: Income Regression Model

```
Call:
lm(formula = median_income ~ policy_score, data = final_data)

Residuals:
   Min     1Q Median     3Q    Max 
-27245 -11662  -3009   8511  44639 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  43102.61    3979.12  10.832  < 2e-16 ***
policy_score   351.05      47.93   7.324 2.98e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 14900 on 71 degrees of freedom
  (2 observations deleted due to missingness)
Multiple R-squared:  0.4304,	Adjusted R-squared:  0.4223 
F-statistic: 53.64 on 1 and 71 DF,  p-value: 2.979e-10
```

Figure 5: Unemployment Regression Model

```
Call:
lm(formula = unemployment_rate ~ policy_score, data = final_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.6515 -1.2873 -0.4435  1.0842  8.2191 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)   6.721516   0.527227  12.749   <2e-16 ***
policy_score -0.005370   0.006351  -0.846    0.401    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.975 on 71 degrees of freedom
  (2 observations deleted due to missingness)
Multiple R-squared:  0.009971,	Adjusted R-squared:  -0.003973 
F-statistic: 0.7151 on 1 and 71 DF,  p-value: 0.4006
```

Figure 6: Poverty Regression Model

```
Call:
lm(formula = poverty_rate ~ policy_score, data = final_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.6839 -3.4615  0.3438  2.8996 13.1329 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  21.21591    1.30774   16.22  < 2e-16 ***
policy_score -0.05215    0.01575   -3.31  0.00147 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.898 on 71 degrees of freedom
  (2 observations deleted due to missingness)
Multiple R-squared:  0.1337,	Adjusted R-squared:  0.1215 
F-statistic: 10.96 on 1 and 71 DF,  p-value: 0.001467
```

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